

- 25** The secondary coil of an ideal transformer delivers an r.m.s. current of 1.5 A to a load resistor of resistance 10 Ω . The r.m.s. current in the primary coil is 5 A.

What is the r.m.s. potential difference across the primary coil?

A 4.5 V

B 6.4 V

C 15 V

D 50 V

Ans: A

$$P = I^2 R = 1.5^2 \times 10 = 22.5 \text{ W}$$

$$V_s = \frac{P}{I_s} = \frac{22.5}{5} = 4.5 \text{ V}$$

$$\text{alt: } V_s = RI = 10 \times 1.5 = 15 \text{ V}$$

$$\frac{V_p}{V_s} = \frac{I_s}{I_p}$$

$$V_p = \frac{1.5}{5} \times 15 \\ = 4.5 \text{ V}$$

- 26** The diagram represents in simplified form some of the energy levels of the hydrogen atom